

MADHAV MENON

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EDUCATION

Manipal Institute of Technology, Manipal, Karnataka

Bachelor of Technology — Computer & Communications Engineering

July 2023 – August 2027

84.7% (8.47/10)

Air Force Golden Jubilee Institute, New Delhi

March 2022 – June 2023

CBSE Class 12 Board Examination

94.6% (473/500)

St. Annes Convent School, Chandigarh

March 2020 – June 2021

CBSE Class 10 Board Examination

96.2% (481/500)

SKILLS

Languages

Python, C, C#, C++, Bash, Typescript, Java, HTML, CSS, Javascript, x86 Assembly

Cryptography

Sagemath, Pycryptodome, Cryptanalysis

Reverse Engineering

Debugging, Ghidra, IDA, GDB, x64dbg, Frida, pwntools, dnSpy

Databases

SQL, PostgreSQL, MongoDB

Graphics Programming

OpenGL, WebGPU

Other Software & Frameworks

Git, Docker, WinAPI, Flask, Linux, Ollama, Wireshark, Android Studio, Unity, Godot

Coursework

Object Oriented Programming, Computer Organization and Architecture,

Embedded System Design, Computer Networks and Protocols

ACHIEVEMENTS

- **Finalist at Smart India hackathon 2025**, India's largest student innovation hackathon. Led a team of 6 members to participate under the problem statement of building an application software to obfuscate the object file using Low Level Virtual Machine (LLVM).
- **Finalist at Cython 2024**, an all India hackathon organized by **NTRO, FITT, Indian Institute of Technology Delhi** for building a parallellised linux kernel fuzzer for increased speed, and code coverage.
- **Awarded First Position and 25,000 INR at Craw CTF 2025**, cybersecurity competition hosted by **Indian Institute of Technology - Jodhpur & sponsored by Craw Security**, as part of IITJ's annual tech fest.
- **Awarded First position and 40,000 INR at Shunya CTF 2024**, a two stage cybersecurity competition conducted by team ncreeps and MIT-ADT University, Pune.

TECHNICAL EXPERIENCE

Cryptography Head

Cryptonite

January 2024 – Present

Manipal, Karnataka, India · On-site

- **Extensively played over 100 CTFs in Cryptography and Reverse Engineering categories**, competing with top teams globally.
- Team is ranked **#2 nationally** as of 21/07/2025 and **finished #3 nationally in 2024** on ctftime.org.
- Created 5 challenges in my domains for **niteCTF 2024**, the team's annual cybersecurity Capture The Flag event, that garnered over 3000 participants globally, & was sponsored by **IITB Trust Lab, Google Cloud Platform**.
- **Conducted a successful Game Hacking Workshop with over 100 attendees at Manipal Institute of Technology**. Taught concepts of debugging, reverse engineering, memory manipulation, & x86 assembly using tools such as x64dbg, cheat engine to freshers, along with live demonstrations.
- Participating in structured team research on post-quantum cryptography, with focus on developing a lattice based bit predictor for the mersenne twister PRNG.

PROJECTS

• **InkSpire** 🌐 🌐

WebGPU, Typescript, Node

<https://github.com/IC3lemon/inkspire>

- GPU-accelerated drawing application built with WebGPU compute shaders and TypeScript
- Implements vector stroke rendering with hardware acceleration, eliminating CPU bottlenecks of traditional canvas APIs
- Currently under development

• **Cryptobible** 🌐

Python, Sagemath

<https://github.com/IC3lemon/cryptobible>

- Python implementations of 60+ cryptographic attacks and utilities.
- Modular architecture with extensive documentation and reusable components.
- Custom python package and CTF tool.

• **Nitefuscator** 🌐

C++, C, LLVM, Capstone, lief, x86 asm, bash, Python

<https://github.com/IC3lemon/Nitefuscator>

- Additional provision to act as a native x86-64 PE ELF binary to binary obfuscator.

- Designed to perform control flow obfuscation, control flow flattening, instruction substitution, general code obfuscation, ROP based obfuscation, anti-debugging.

- **fuzzez**

Python, Ansible, Syzkaller, bash, PyQt5

- An attempt at a large scale parallelised linux kernel module fuzzer.
- Designed to speed up the process of fuzzing and increase code coverage.
- Placed 2nd at Cython 2024, IIT-Delhi (hackathon organised by NTRO x FITT)